

Design FIR filter

step 1: calculate δ_1 ; δ_2 ; ω_p ; ω_s

step 2: calculate Transition width, $\Delta\omega$:

$$\Delta\omega = \omega_s - \omega_p$$

step 3: Find Minimum value of ripple, δ :

$$\delta = \min(\delta_1, \delta_2) =$$

step 4: calculate Attenuation, A :

$$A = -20 \log_{10}(\delta) \text{ dB}$$

step 5: calculate cut-off frequency; $\omega_c = \frac{\omega_s + \omega_p}{2}$

step 6: Calculate β :

$$\beta = \begin{cases} 0.1102 (A - 8.7) & ; A > 50 \\ 0.5842 (A - 21)^{0.4} + 0.07886 (A - 21) & ; 21 \leq A \leq 50 \\ 0 & ; A < 21 \end{cases}$$

step 7: Calculate M :

$$M = \frac{A - 8}{2.285 \Delta\omega} \quad \text{and} \quad \alpha = \frac{M}{2}$$

step 8: For Kaiser window:

$$w(n) = \begin{cases} \frac{I_0 \left\{ \beta \left(1 - \left(\frac{n-\alpha}{\alpha} \right)^2 \right)^{\frac{1}{2}} \right\}}{I_0 \beta} & ; 0 \leq n \leq M-1 \\ 0 & ; \text{otherwise} \end{cases}$$

Bilinear Transformation

step 1: Determine design specifications;
 A_p ; A_s ; ω_p ; ω_s ;

step 2: Obtain Ω_s ; Ω_p ;

for BLT: $\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right)$ (for Ω) For I.I.M; $\omega_s = \frac{\omega_c}{T}$

step 3: obtain order of filter, N ; for A_s & A_p are not in dB:

$$\text{For } A_s \text{ \& } A_p \text{ in dB } N = \frac{1}{2} \left[\frac{\log\left(\frac{10^{0.1 A_s} - 1}{10^{0.1 A_p} - 1}\right)}{\log\left(\frac{\Omega_s}{\Omega_p}\right)} \right] \quad N = \frac{1}{2} \left[\frac{\log\left[\frac{\frac{1}{A_s} - 1}{\frac{1}{A_p} - 1}\right]}{\log\left(\frac{\Omega_s}{\Omega_p}\right)} \right]$$

step 4: Determine Cut-off frequency: ω_c ;

$$\Omega_c = \frac{\Omega_p}{\left[10^{0.1 A_p} - 1\right]^{1/2N}}$$

step 5: Determine pole location;

$$P_i = \pm \Omega_c e^{j(N+2i+1)\pi/2N} ; i = 0, 1, \dots, N-1$$

step 6: Determine Transfer function;

$$H_a(s) = \frac{\Omega_c^N}{(s-s_1)(s-s_1^*)(s-s_2)(s-s_2^*)}$$

for BLT
put $s = \frac{2}{T} \left(\frac{z-1}{z+1}\right)$

For stability of filter,

~~put~~ p_1, s_1, s_2, s_2^* are the poles lies in L.H.S of s plane

S.N.	Window	Time-Domain sequence
1.	Rectangular window $A_s = -21 \text{ dB}$	$w_R(n) = \begin{cases} 1 & ; 0 \leq n \leq M-1 \\ 0 & ; \text{otherwise} \end{cases}$
2.	Hamming window $A_s = -53 \text{ dB}$ $A_s = -44 \text{ dB}$	$w_{Hm}(n) = \begin{cases} 0.54 - 0.46 \cos \frac{2\pi n}{M-1} & ; 0 \leq n \leq M-1 \\ 0 & ; \text{otherwise} \end{cases}$
3.	Hanning window -44 dB $A_s = (53 \text{ dB})$	$w_{Hn}(n) = \begin{cases} \frac{1}{2} \left(1 - \cos \frac{2\pi n}{M-1} \right) & ; 0 \leq n \leq M-1 \\ 0 & ; \text{otherwise} \end{cases}$
4.	Kaiser window	$w_K(n) = \begin{cases} \frac{I_0 \left\{ \beta_0 \left(1 - \left(\frac{n-\alpha}{\alpha} \right)^2 \right)^{\frac{1}{2}} \right\}}{I_0 \beta} & ; 0 \leq n \leq M-1 \\ 0 & ; \text{otherwise} \end{cases}$

step 1: Obtain $h_d(n)$;

$$h_d(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} H_d(\omega) d\omega$$

step 2: obtain $h(n)$;

$$h(n) = h_d(n) \cdot w(n)$$

step 3: Calculation $h(n)$;

Hanning window

Hamming window

width of main lobe

$$= \frac{8\pi}{M}$$

$$\therefore K \left(\frac{8\pi}{M} \right) = \frac{8\pi}{M} \Rightarrow K=1$$

$$N = K \left(\frac{2\pi}{\omega_2 - \omega_1} \right)$$

Z-Transform

SN.	x	Z-Transform	ROC
1.	1	1	Entire z-plane
2.	$\delta(n)$	z^{-k}	Entire z-plane Except $z=0$
3.	$\delta(n-k)$	z^k	Entire z-plane, except $z=\infty$
4.	$u(n)$	$\frac{1}{1-z^{-1}} = \frac{z}{z-1}$	$ z > 1$
5.	$u(n)$	$\frac{z^{-1}}{z^{-1}-1}$	$ z > 1$
6.	$nu(n)$	$\frac{z}{(z-1)^2}$	$ z > 1$
7.	$a^n u(n)$	$\frac{1}{1-az^{-1}} = \frac{z}{z-a}$	$ z > a$
8.	$na^n u(n)$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z > a $
9.	$-a^n u(n-1)$	$\frac{1}{1-az^{-1}}$	$ z < a $
10.	$-na^n u(n-1)$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
11.	$(\cos \omega_0 n) u(n)$	$\frac{1-z^{-1}\cos \omega_0}{1-2z^{-1}\cos \omega_0+z^{-2}}$	$ z > 1$
12.	$(\sin \omega_0 n) u(n)$	$\frac{z^{-1}\sin \omega_0}{1-2z^{-1}\cos \omega_0+z^{-2}}$	$ z > 1$
13.	$(a^n \cos \omega_0 n) u(n)$	$\frac{1-az^{-1}\cos \omega_0}{1-2az^{-1}\cos \omega_0+a^2z^{-2}}$	$ z > a $
14.	$(a^n \sin \omega_0 n) u(n)$	$\frac{az^{-1}\sin \omega_0}{1-2az^{-1}\cos \omega_0+a^2z^{-2}}$	$ z > a $

$\frac{d}{dz} \left(\frac{1}{1-az^{-1}} \right) \rightarrow -z \frac{d}{dz} \left(\frac{1}{1-az^{-1}} \right)$
 Differentiation property
 $nx(n) \rightarrow -z \frac{d}{dz} X(z)$
 $\frac{d}{dz} \left(\frac{1}{1-az^{-1}} \right) \rightarrow \frac{z}{(z-1)^2}$
 Scaling property